

Zihao Chen

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Education

The University of Chicago <i>B.S. in Computer Science; B.S. in Neuroscience (GPA: 3.98/4.0)</i> • Courses: AI Agent, Machine Learning, Software Development, System Programming, Computer Security • Honors: Phi Beta Kappa Academic Honor Society, Enrico Fermi Scholar (Top 5%), Janet Rowley Scholar (Top 5%)	Chicago, IL 06/2026
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Research Experience

Research Assistant <i>Hatsopoulos Lab - University of Chicago</i> • Engineered a fault-tolerant, modular, and extensible Python pipeline to stream, synchronize, and store real-time IoT data at high throughput (~80 GB/session at 200 MBps). • Developed a computer vision and data processing workflow using OpenCV and PyTorch to perform feature extraction and noise reduction on large-scale datasets. Implemented data filtering and Principal Component Analysis (PCA) to reduce feature dimensions from 256 to 25 while improving model prediction accuracy by 15%. • Trained and evaluated multiple machine learning models (LDA, SVM, Kalman Filter) for time-series prediction, systematically iterating on model selection to enhance predictive performance and robustness against outliers. • Presented research findings at multiple symposia and a major international conference (Society for Neuroscience).	Chicago, IL 06/2024 – now
Research Assistant <i>Attention, Perception, & Experience (APEX) Lab - University of Chicago</i> • Optimized a Neuro Evolution of Augmenting Topologies (NEAT) program by incorporating Compositional Pattern-Producing Networks (CPPN), significantly boosting performance and the complexity of the output. • Developed automated data validation and cleaning scripts using Pandas for large Electroencephalogram (EEG) datasets, reducing manual data correction effort by 80% and improving downstream model performance.	Chicago, IL 09/2023 – 05/2024

Teaching Experience

Teaching Assistant (Mathematical Foundation of Machine Learning) • Led discussion sections for 23 students, covering topics in linear algebra, probability, and machine learning algorithms. • Designed and implemented weekly code demos and rigorous assignments for 150+ master's and undergraduate students, directly impacting the computational and theoretical skill development of the cohort. • Provided one-on-one mentorship during weekly office hours, effectively debugging student code and clarifying theoretical concepts for successful assignment completion and deep learning comprehension.	09/2025 – now
Teaching Assistant (Introduction to Computer Science) • Evaluated Python assignments for 20-25 students weekly, providing targeted feedback on software development. • Collaborated with professors to refine grading rubrics and course materials based on student performance trends.	01/2024 – 06/2025

Presentations

• Chen, Z., Sobinov, A., Raman, C., & Hatsopoulos, N., (2025, November 13). <i>Encoding of grasp force in human motor cortex during object transport</i> [Paper presentation]. Society for Neuroscience 2025, San Diego, CA, United States.
• Chen, Z., Sobinov, A., Raman, C., & Hatsopoulos, N., (2025, April 25). <i>Exploring missing grasp force signals in motor cortex during object transport</i> [Poster session]. QUAD Research Symposium, Chicago, IL, United States.
• Chen, Z., Sobinov, A., Raman, C., & Hatsopoulos, N., (2024, August 13). <i>Decoding grasp force in the M1 during object transportation</i> [Paper presentation]. Jeff Metcalf's Research Fellowship Symposium, Chicago, IL, United States.

Selected Research Projects

AI-Powered Resume Creation and Optimization Application • Architected an AI-powered application to automate the resume lifecycle, from data ingestion (.csv or form inputs) and multi-version LaTeX rendering to AI-driven resume analysis and content generation. • Engineered a custom framework to manually implement tool calling based on structured output , enabling LLMs to perform live web scraping, verify data via external APIs, and interact with the local file system. • Benchmarked Qwen 2.5 32B against Llama 3 70B and GPT-3.5, selecting it for its superior native support for structured output and efficient performance on consumer 24GB GPUs. • Leveraged the GPT-4 API with prompt engineering and tool calling to increase high-impact action verbs by 50% and boost resume quality scores by 35% based on an internal rating model.	06/2024 – 08/2024
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